Salifort Motors HR Analytics

Predictive Employee Churn Modeling & Data-Driven Retention Strategies

Business Case & Problem

Understanding employee turnover patterns to protect institutional knowledge and reduce costly hiring cycles.

The Cost of Turnover

- **High Recruitment Expenses:** Finding, interviewing, and onboarding replacement talent requires heavy capital investment.
- **Loss of Productivity:** Teams face increased strain during gap periods while new employees ramp up.
- **Predictive Intelligence:** Identifying flight-risk employees early allows HR to execute targeted retention interventions.
- **Data-Driven Approach:** Transitioning from reactive offboarding to proactive employee satisfaction management.

Data Pipeline & Cleaning

Standardization & Deduplication

Cleaned 11,991 unique records by removing exact duplicates and standardizing feature column titles to lowercase snake_case (e.g., average_hours_per_month and total_time_spend).

Feature Engineering & Encoding

Mapped categorical salary levels into ordinal tiers (low: 0, medium: 1, high: 2) and created dummy variables for departments to prepare seamless inputs for tree-based estimators.

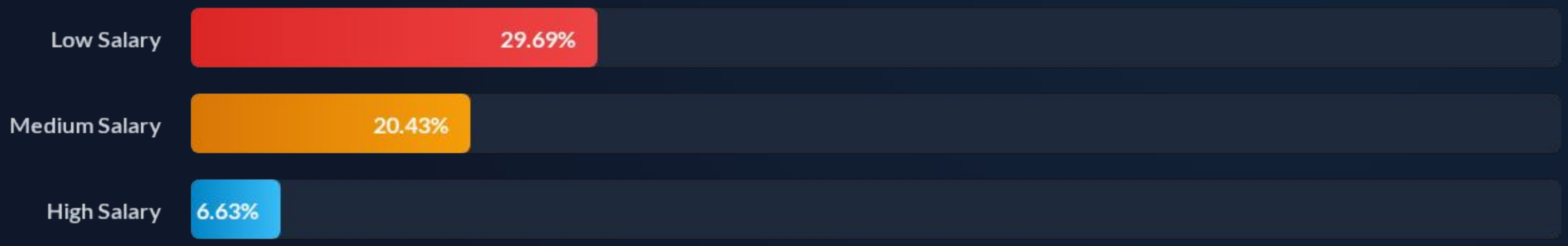
Exploratory Insights

Overwork drives turnover

EDA revealed clear employee clusters: staff assigned to 6–7 projects worked over 240+ hours per month, suffering extreme burnout and high churn rates.

Conversely, staff assigned to only 2 projects suffered from under-utilization, showing lower satisfaction and elevated departure risk.

Churn Rate by Salary Tier



Employees in the lower salary bracket experience nearly 4.5x higher churn compared to high-earning staff, pointing to financial compensation and promotion access as major retention anchors.

Baseline Attrition Rate

16.6%

Overall Turnover Rate

Substantial Attrition Signal

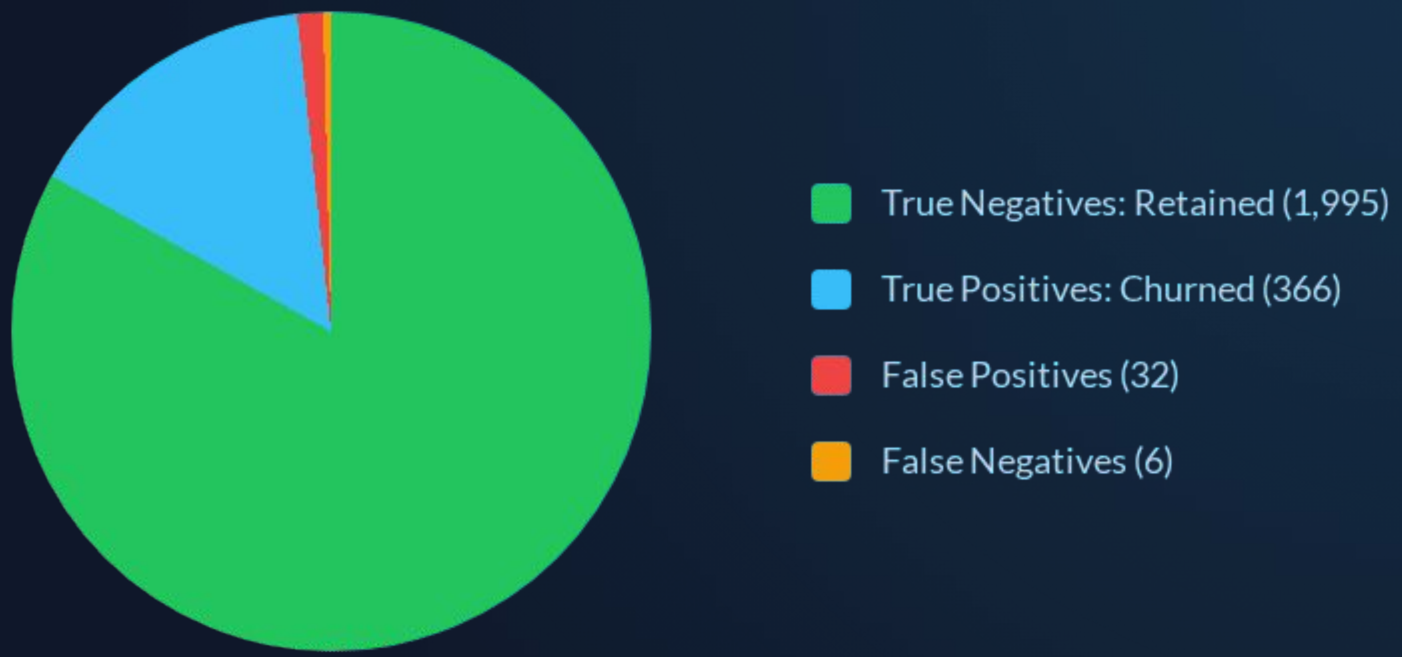
Out of 11,991 total employees evaluated after deduplication, 1,991 employees left the company. This baseline imbalance highlights the need for precision-focused modeling to accurately capture churn risks without excessive false positives.

Model Benchmark Comparison

Model Classifier	Accuracy	Precision	Recall	F1-Score
Tuned Decision Tree	97.99%	97.98%	97.99%	0.9797
Random Forest (Winner)	**98.38%**	**98.38%**	**98.38%**	**0.9836**
XGBoost Classifier	98.16%	98.16%	98.16%	0.9813

Evaluated using 5-fold cross-validation grid search. Random Forest achieved the highest F1-score and generalization stability across unseen test sets.

Test Matrix Breakdown



Test Set Confusion Matrix: The Random Forest model achieved an exceptional 99.7% Recall on holdout test data, missing only 6 churned employees out of nearly 2,400 test cases.

Top Drivers of Turnover

Satisfaction Level

The single most dominant feature (~46% relative importance). Employees with satisfaction below 0.12 present immediate flight risk.

Project Contribution

Second highest driver (~20% importance). Employees assigned to 6+ projects experience extreme dissatisfaction and workload fatigue.

Monthly Hours & Tenure

Working over 200 hours per month combined with 4–5 years of tenure without promotion creates severe turnover vulnerability.

Strategic HR Initiatives

- **Cap Maximum Workload:** Limit project assignments to a maximum of 5 projects and cap monthly work hours at 220 hours to prevent burnout.
- **Support Under-Utilized Staff:** Re-engage staff assigned to only 2 projects with mentoring or additional growth opportunities.
- **Evaluate Promotion Access:** Review compensation structures and growth pathways for employees with 3+ years of tenure in low/medium salary brackets.
- **Proactive Satisfaction Tracking:** Deploy regular, anonymous pulse surveys to trigger early retention interventions when satisfaction drops below 0.40.